

**RiceDoc: A Mobile-Based Rice Leaf Disease Detection Through Image
Recognition of Rice Leaf Images using Convolutional Neural Network**



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CHAPTER 1

INTRODUCTION

Background of the Study

Rice is a staple food crop and one of the most widely grown and consumed crops in the world. The rice is susceptible to a range of diseases that can have a significant impact on its growth and yield. As one of the most destructive diseases. In the Philippines, there is a big demand for this crop because the average diet is based on rice [1]. In 2013 alone, the domestic consumption for milled rice in the country was reported to be 12.85 million metric tons. The rate of growth of rice production, however, has slowed down [2]. Rice crop loss is now one of the major concerns in rice production. Crop loss has become a great risk particularly for poor farmers with unfavorable environments, and on food security in general [3]. Rice crop disease may affect the growth of the rice crop, especially to the farmers.

Rice Leaf Blast or Blast is caused by the fungus *Magnaporthe Oryzae*. This disease occurs in areas with low soil moisture, frequent and prolonged periods of rain shower, and cool temperature in the daytime [4]. Reportedly, between 10% and 30% of the annual rice harvest is lost due to infection by the rice blast fungus *Magnaporthe oryzae* [5]. Brown spot is a fungal disease caused by the fungus *Cochliobolus miyabeanus* that infects the coleoptile or the protective sheath covering the leaf, and glumes or the outer husk surrounding the grain. This disease develops in areas with high relative humidity and temperature between 16 and 36°C [6]. Sheath Blight is caused by a fungal disease called *Rhizoctonia solani*. The infected leaves dry out and die more rapidly. This occurs in areas with high temperature (28-32°C), high levels of nitrogen fertilizer, and relative humidity of crop canopy from 85-100% [7].

The Davao region is one of the country's key rice-producing areas, contributing considerably to the national rice supply [8][9]. Nevertheless, rice leaf diseases such as blast, bacterial leaf blight, and sheath blight can cause considerable yield losses, lowering rice production productivity and profitability.

Considering the importance of rice leaf diseases in the region, there is little information on the yield losses caused by these diseases [9].

Image processing is a useful method for detecting diseases in rice plants that has piqued the interest of agricultural researchers and professionals. Images captured by a digital camera, were processed using image growing, image segmentation techniques to detect infected parts of the plants [10]. Then the infected leaf is being used for classification using the proposed neural network [10]. Furthermore, image processing can be used to identify infections even before the symptoms are visible to the naked eye, allowing for earlier intervention and epidemic prevention.

Rice is a vital crop worldwide, providing a staple food source for a significant portion of the global population. However, rice production is severely affected by various diseases that affect the health and yield of rice plants. This is why the researchers aims to develop the RiceDoc: A Mobile-based Rice Leaf Disease Detection Through Image Recognition of Rice Leaf Images. This application helps the farmers to identify what kind of disease in their crop and also provide information of the disease and its prescribed medications that may help to avoid the spreading of the disease. This study aims to assess the effectiveness of the application as a tool in farming.

Objectives of the Study

This study aims to develop an Android application capable of detecting rice diseases using the rice plant's leaves. Specifically, it refers to:

1. Identify diseases of rice leaves using image processing techniques, for rice leaf blast, brown spot, sheath blight that uses the Convolutional Neural Network algorithm architecture which is the DenseNet121.
2. Provide information of the detected disease, commonly prescribed preventive measures.

3. Compute the impact of the disease on its production and estimated damages if the disease is present in terms of the land area coverage.
4. Predict the harvest or production output of the crop yield using data mining technique particularly linear regression, and provide graphical visualization using graph to the output.

Significance of the Study

This section of the document discusses the benefits that they will get by the following:

Economic Growth. Detection of rice leaf diseases significantly helps the economic growth, specifically in rice production.

Rice Farmers. Early detection of the identified disease helps farmers to mitigate the crop yield loss due to the disease.

City Agriculture Office. With regards to the rice focal of the said department, the proposed system will help the experts to educate and guide rice farmers for better crop yield progress.

Future Developers. For the future developers, this study could be used as reference in developing applications with machine learning approach.

Scope and Limitation

The focus of the study is to create a mobile application which will detect rice diseases through its leaves and it will run only on Android devices. The study includes image processing, which allows for the classification of healthy and sick leaves, the advantage of image-based disease identification is that it allows for large-scale crop monitoring, which is especially essential in the case of rice, a staple commodity that feeds billions of people throughout the world.

After detecting the disease and giving information as well as commonly prescribed medications, it will also provide the estimated damages if the disease is present in a particular yield. Image classification model will be evaluated its performance through performance metrics. Also, it will feature forecasting that allows farmers and experts to predict their harvest or production output given by the factors that affects the its production. However, the study limits only on three common rice leaf diseases; other rice leaf diseases such as False smut, grassy stunt, and several others are not included in this study.

Review of Related Literature and Works

This section discusses the different sources of literatures and works which contribute to the crafting of this research that is to be presented.

Crop yields and Diseases

Rice leaf blast, also known as blast, holds significant economic importance and has the potential to cause substantial yield losses in rice, ranging from 70% to 80% [11]. The traditional Jumli Marshi variety in Jumla, Karnali, is highly susceptible to blast, as evidenced by the destruction of 80% of Jumli Marshi crops in the Tila valley of Jumla in 2011 [12]. Additionally, between 2000 and 2004, blast epidemics affected a vast area of rice fields, estimated at 3.3 to 5.7 million hectares, in southeastern and central China along the Yangzi River Delta [13]. In Sichuan, the most significant rice-producing province in China, leaf and panicle blast affected 0.5 million and 0.3 million hectares of rice fields, respectively, resulting in a yield loss of 400 thousand tons in 2005 alone [13]. Furthermore, in 2004, rice blast caused a 70% reduction in total yield in upland rice production in West Java, Indonesia [14].

Brown spot is of great importance in several countries and has been reported to cause enormous losses in crop yield, reaching up to 90%. This is particularly evident when the leaf-spotting phase of the disease reaches epidemic proportions, as observed during the Great Bengal Famine in 1942 [15]. The Great

Bengal Famine, which occurred in India during 1942-1943, was primarily attributed to a widespread outbreak of brown spot, leading to yield reductions of approximately 80% [16]. Similarly, in Nigeria, it has been estimated that severe brown spot infections caused a yield reduction of 30-43% [17].

Sheath blight has become increasingly significant in terms of its economic impact in the past two decades [18]. In Japan, an estimated 120,000 to 190,000 hectares of rice fields have been affected by sheath blight, resulting in an annual loss of approximately 24,000 to 38,000 tons of rice [19]. In one district of Sri Lanka, it was found that sheath blight prevalence in rice fields affected around 10% of the rice tillers [20]. Meanwhile, data from mainland China suggests that the average losses due to sheath blight were 12.6% in 1980 and 9.1% in 1981 [21].

Rice leaf blast, brown spot, and sheath blight are major diseases of rice in the Philippines that have caused severe yield losses [22]. Rice leaf blast caused extensive losses in the Philippines during 1962 and 1963, with certain regions experiencing yield reductions exceeding 90%. Specifically, the provinces of Bicol and Leyte encountered losses ranging from 50% to 60% across thousands of hectares cultivated with the Peta variety [23]. Another blast outbreak in Laguna and Quezon from 1969 to 1970 resulted in yield losses of BPI-76 and C4-63 cultivars ranging from 70% to 85% [24]. Brown spot disease was found to occur in all rice fields in Zamboanga City, Philippines. The severity of infection ranged from 25% to 75% and above. Severe brown spot disease outbreak occurred along the western coast during the dry season. Furthermore, the cultivated rice varieties were found to be susceptible to brown spot disease throughout all stages of growth [25]. Sheath blight has caused yield reductions ranging from 5% to 80% in rice grain production in the Philippines. The associated yield losses per hectare range from 0.27 to 1.29 tons during the dry season and from 0.23 to 1.37 tons during the wet season [26][27].

Image Processing

Image processing techniques have been widely employed in agricultural research, including disease detection in crops. A number of disease detection,

identification and qualification methods have been developed and applied in a wide variety of crops. These techniques enable the extraction and analysis of relevant features from images to identify and classify diseases accurately. Image segmentation, feature extraction, and image enhancement methods have been utilized to preprocess rice leaf images, improving the effectiveness of subsequent disease detection algorithms [28].

Convolutional Neural Network Algorithm

In this study, the algorithm used was a convolutional neural network. Convolutional neural networks have been widely employed in the field of image processing as high-performing algorithms that produce outstanding results by utilizing their own local receptive fields, weight sharing, pooling, and sparse connections. The convolutional neural network (CNN), one of the more advanced neural network techniques, has a useful model class for interpreting handwritten text. The neural network can be trained to learn more and execute complex tasks with high precision since it operates similarly to the human brain. This technique can be used for text and damaged documents since it analyzes each image pixel and compares it to data that has already been gathered. When compared to other approaches, neural networks are now the most successful in resolving problems, such as identifying visually trending patterns or stock market data [29].

In 2021, the study entitled Classification of rice plant diseases using the convolutional neural network method [30] used the Convolutional Neural Network (CNN) to classify the types of rice disease. The model was trained on a dataset of 105 images, which achieved an accuracy of 95.24%. Similar study entitled Rice leaf disease classification using CNN [31] used a deep learning-based approach to detect and classify rice diseases from images. The study aids farmers in obtaining a healthy crop yield by identifying illnesses in rice leaves. When compared to machine learning techniques, deep learning models perform better. Specifically, the 5-layer convolution model achieved the highest accuracy of 78.2%, while other models only achieve 58.4%. Also, the study titled Detection of rice plant diseases using convolutional neural network [32] a deep learning-based

model was developed to identify four types of rice diseases using leaf images. The model was trained on a dataset of 2,239 with each type of diseases. The model achieved an accuracy of 91% in detecting the images.

Classification Metrics

Classification metrics play a crucial role in evaluating the performance of machine learning models, including those used in disease detection. Several metrics are commonly used to assess classification accuracy. Accuracy, defined as the ratio of correctly classified samples to the total number of samples, is a widely adopted metric. However, accuracy alone may not be sufficient when dealing with imbalanced datasets, where the number of samples in different classes varies significantly. Precision, recall (sensitivity), and F1-score are metrics that provide a more comprehensive evaluation of classification performance, especially for imbalanced datasets. Precision calculates the proportion of correctly classified positive samples, while recall measures the proportion of actual positive samples correctly identified. F1-score combines both precision and recall into a single metric, considering their harmonic mean [33].

Table 1. Literature Table

Article	Functionalities	Algorithm	Parameters	Performance Metrics
<i>Classification of rice plant diseases using the convolutional neural network method</i> [30]	Image-based classification of diseases	Convolutional Neural Network (CNN)	No. of Epochs = 150, Epoch variations and data augmentation	Accuracy (95.24%)
<i>Rice leaf disease classification using CNN</i> [31]	Image-based detection and classification of diseases	5-layer Convolutional Neural Network	Learning rate, number of epochs, optimizer	Accuracy (78.2%)
<i>Detection of rice plant diseases using</i>	Image-based detection of disease	Convolutional Neural Network	No. of epochs = 10, optimizer, validation	Accuracy (91.41%)

<i>convolutional neural network</i> [32]			split, batch size, metrics, loss	
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The Image Recognition System for Crop Disease Identification of Paddy fields of Sri Lanka [34], is about identifying paddy disease leaf in Sri Lanka with the use of image recognition. Segmentation algorithm, was the approach used in image processing. The results were promising as the classification accuracy was over 70% around 50 images were used to analyze. The researcher's goal in this study was to develop an image recognition system that could recognize paddy leaf diseases, and to automate the classification and recognition of the diseases.

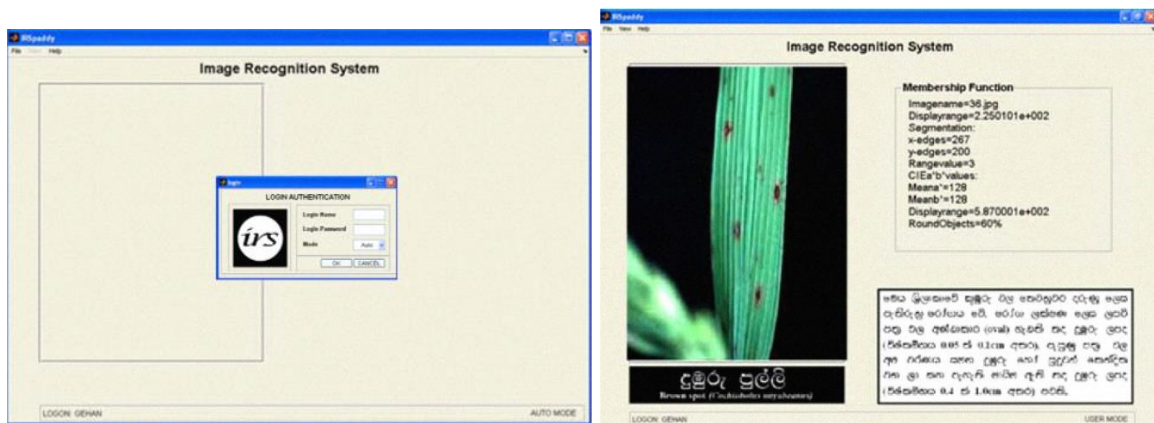


Figure 1. Sample Image for Paddy Image Recognition System on Desktop Application

The Multifunctional Platform and Mobile Application for Plant Disease Detection [35] is a mobile platform for plant disease detection that uses image processing techniques. K-Nearest Neighbor was being used in image classification algorithm of the study; the algorithm is tested through a series of sessions to obtain a better accuracy. Upon reading the study, they trained again using Adam optimizer, an extended version of stochastic gradient descent that are implemented in deep learning which they obtain for 100 epochs in which they have obtained a 95.71% test accuracy on classification.



Figure 2. Sample Images of Platform for Plant Disease Detection

The iDahon: An Android Based Terrestrial Plant Disease Detection Mobile Application Through Digital Image Processing Using Deep Learning Neural Network Algorithm [36] focuses on detection of common terrestrial plants found in the Philippines with the use of image processing through Deep Learning Neural Network. The researcher's aim of this study is to prevent and raise awareness on common diseases of the terrestrial plants, this shows the capability of android based mobile application in detecting or image recognition. The result of the developed application has an accuracy of 80% rate.

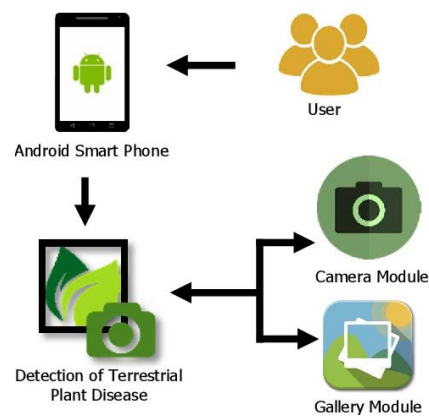


Figure 3. Sample Image for iDahon's Conceptual Framework

Table 2. Comparison of Related Systems

List of Functions	Image Recognition System	PDDP	iDahon:	RiceDoc:
Image-based Search	✓	✓	✓	✓
k-Nearest Neighbor Algorithm		✓		
Convolutional/Deep Learning Neural Network			✓	✓
Segmentation Algorithm	✓			
Offline Application	✓		✓	✓
Online Application				✓
Measures impact of the disease and estimated damages				✓

Synthesis

With the existing literatures presented, table 1 has demonstrated the potential of machine learning algorithms, particularly CNN, for rice disease detection. These studies have shown that CNN-based approaches can achieve high levels of accuracy in detecting and classifying rice diseases. In addition, several studies have emphasized the importance of using a large and diverse dataset for training the machine learning algorithm, as this can improve the algorithm's accuracy and robustness. The use of image-based classification has shown promising results in detecting and classifying rice diseases accurately. Table 2 shows the related works; the proponents are able to identify a few similarities with the application that is being proposed in this study. **Image Recognition System** which has the same function of image recognition and made

the application offline, the downside of this study is that the application was based on desktop and desktop application are not very convenient at all times especially for the farmer's side, they need to purchase computer in order to use the application. That is why the **RiceDoc** wants to develop this application through android platforms similar to **iDahon** and **PDDP**. The **Image Recognition System**, **PDDP**, and **iDahon** only displays the information of the detected disease and display its dimensions of the disease, in **RiceDoc**'s case, it does not only limit on displaying information and prescription of common medications, but also providing the impact of the detected disease and effect to its surrounding.

Definition of Terms

This section defines the following terms that are used in this research;

Accuracy.	A performance metric in machine learning to determine the performance of the classification model.
City Agriculture Office.	An LGU sector which helps the local farmers to produce, export and monitor good quality crops.
F1 Score.	A performance metric that combines precision and recall into a single metric to measure the classification model's performance.
Image Classification.	Uses machine learning algorithm to analyze the picture and categorizing each picture.
Neural Network.	A method that is used in Artificial Intelligence that enables computers to understand the process of data.
Precision.	A performance metric in machine learning that evaluates the positive predictions made by the classification model.

Recall.

A performance metric in machine learning that measures the ability of a classification model to identify positive instances.

Rice Farmer.

A farmer specifically in rice, that produces and harvests rice crop yield.

CHAPTER 2

METHODOLOGY

This chapter discusses the methods undertaken by the proponents for the development of the project. Agile Methodology was applied in developing this project. One of the approaches used in software development is agile software development. Agile refers to being quick, light, mobile, and aware. A process model notion that differs from the ones currently in use is referred to as being agile [37].

Figure 4 illustrates the agile software development methodology which proponents have followed the phases involved in the System Development Life Cycle (SDLC) such as system planning, system analysis, system designing, system testing and implementation, and system maintenance of the project.

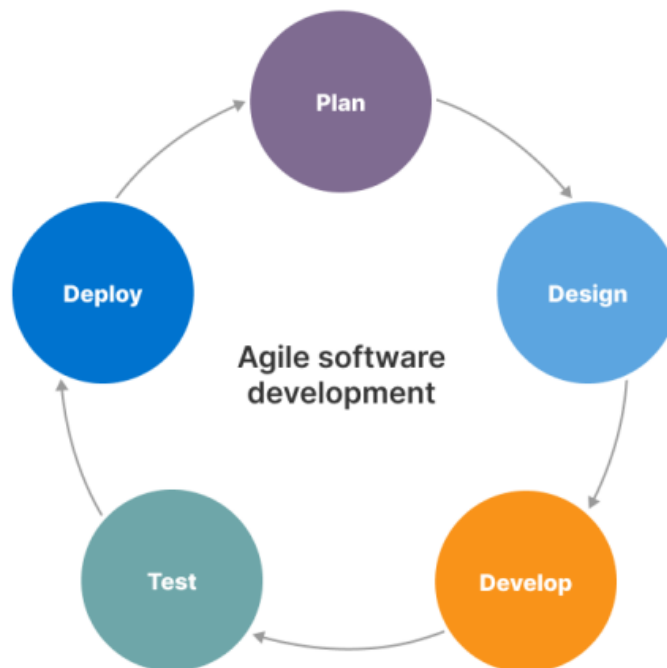


Figure 4. Agile Software Development Methodology

System Planning

The proponents presented three probable titles to the panel during preliminary defense. After the presenting the prospected titles to the panel members, the panel have agreed that the first title presented which is RiceDoc: A Mobile-Based Rice Leaf Disease Detection Through Image Recognition of Rice Leaf Images using Convolutional Neural Network was chosen to be the official capstone project of the proponents.

The proponents conducted an environmental scanning last March 01, 2023 at City Agriculture Office of the City Government of Panabo, to gain insights and information about the common rice leaf diseases that are currently present in the rice farmlands of Panabo City and what are tools do they use to detect each leaf diseases. The proponents created a project to provide an easy solution in aiding the rice farmers to detect the rice leaf diseases at early stage through their mobile devices.

Project Organization

Each member of the project development team has an individual role and function that contribute in this project. Figure 5 shows the project team organization and the designated individual member of the project team.

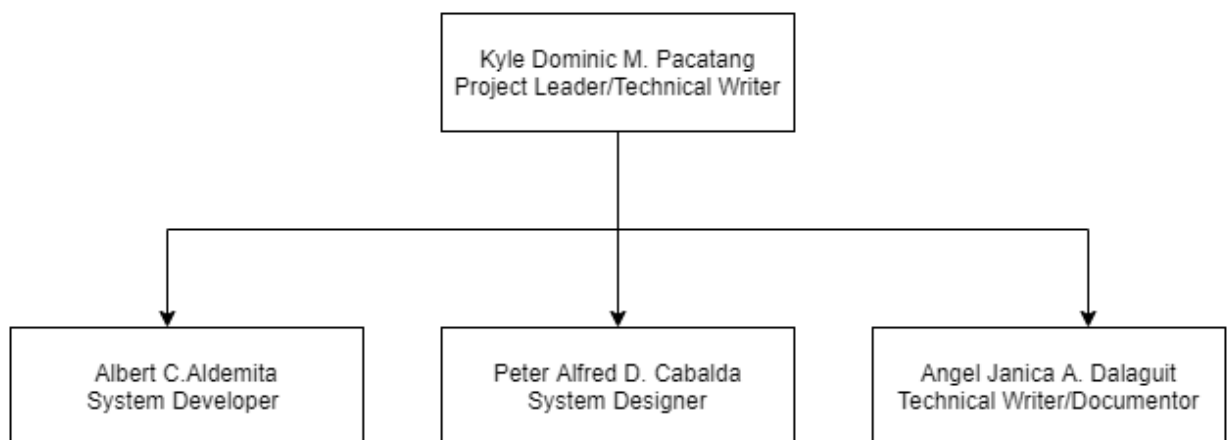


Figure 5. Chart of the Project Development Team

Work Breakdown Structure

This section elaborates the work breakdown structure of the development phases as followed by the proponents. A work breakdown structure shows the project's deliverables, tasks, and sections in a structured way. Figure 6 illustrates the project that is being divided into manageable parts, giving a clear and organized picture of the labor necessary to accomplish project goals.

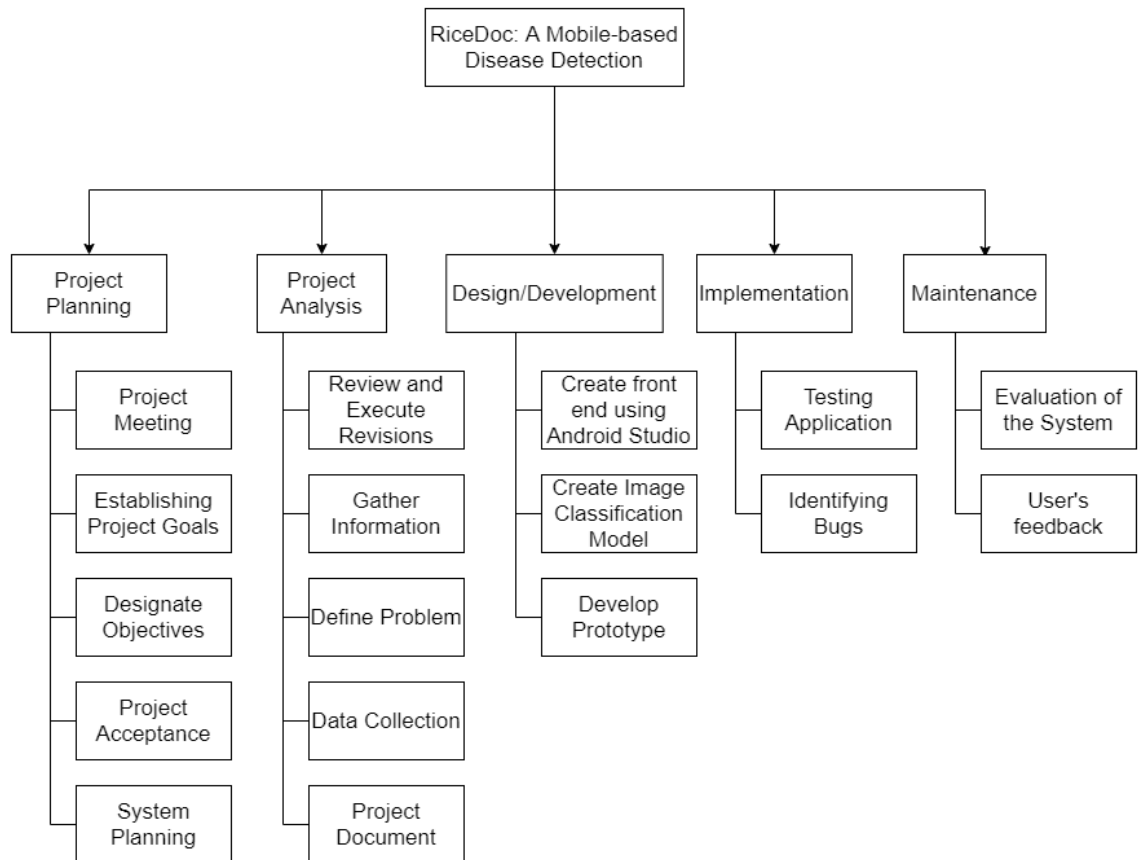


Figure 6. Work Breakdown Structure of the Project

Gantt Chart

This section presents the project schedule for the development of RiceDoc: A Mobile-based Rice Leaf Disease Detection Through Image Recognition of Rice Leaf Images using Convolutional Neural Network in Figure 7 as listed in the Work Breakdown Structure. A Gantt Chart is a bar chart that illustrates the project duration in months or weeks.

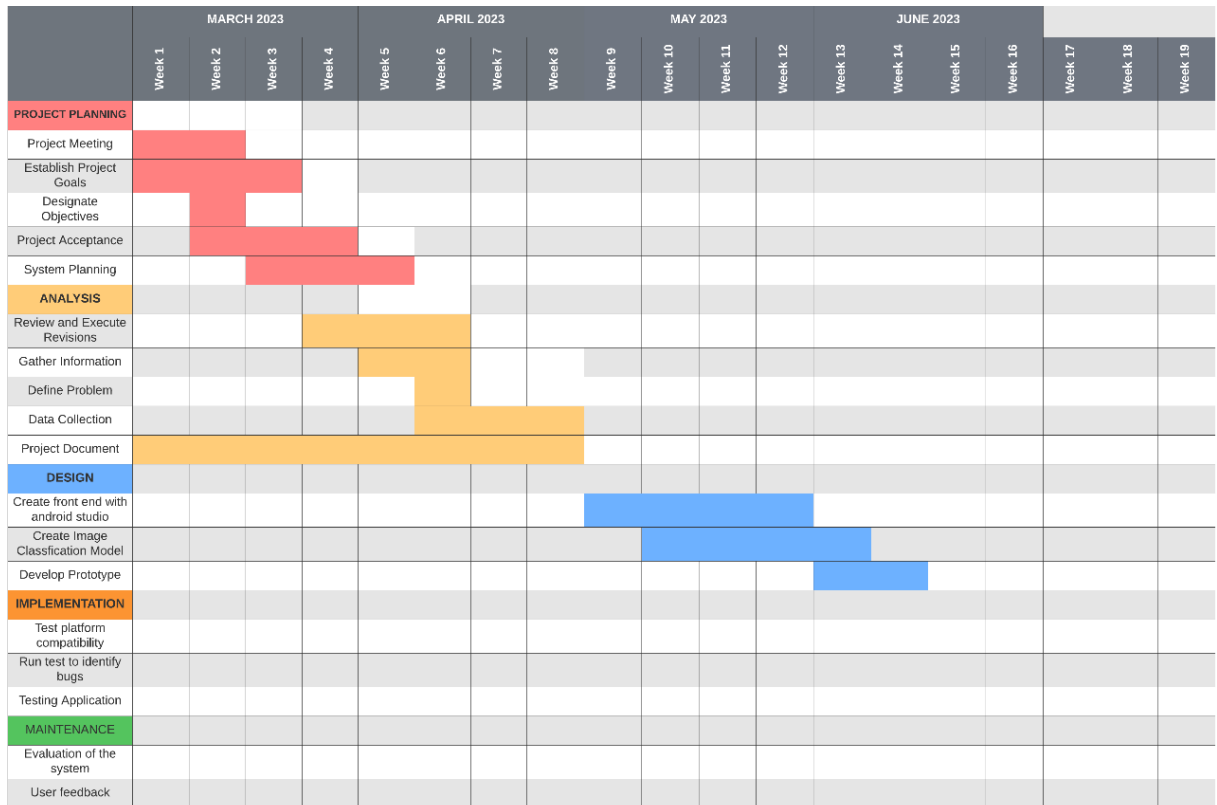


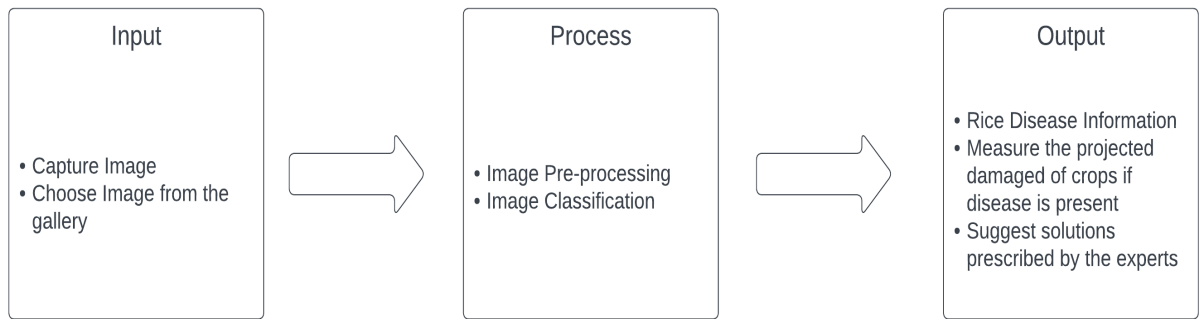
Figure 7. Gantt Chart of the Project

System Analysis

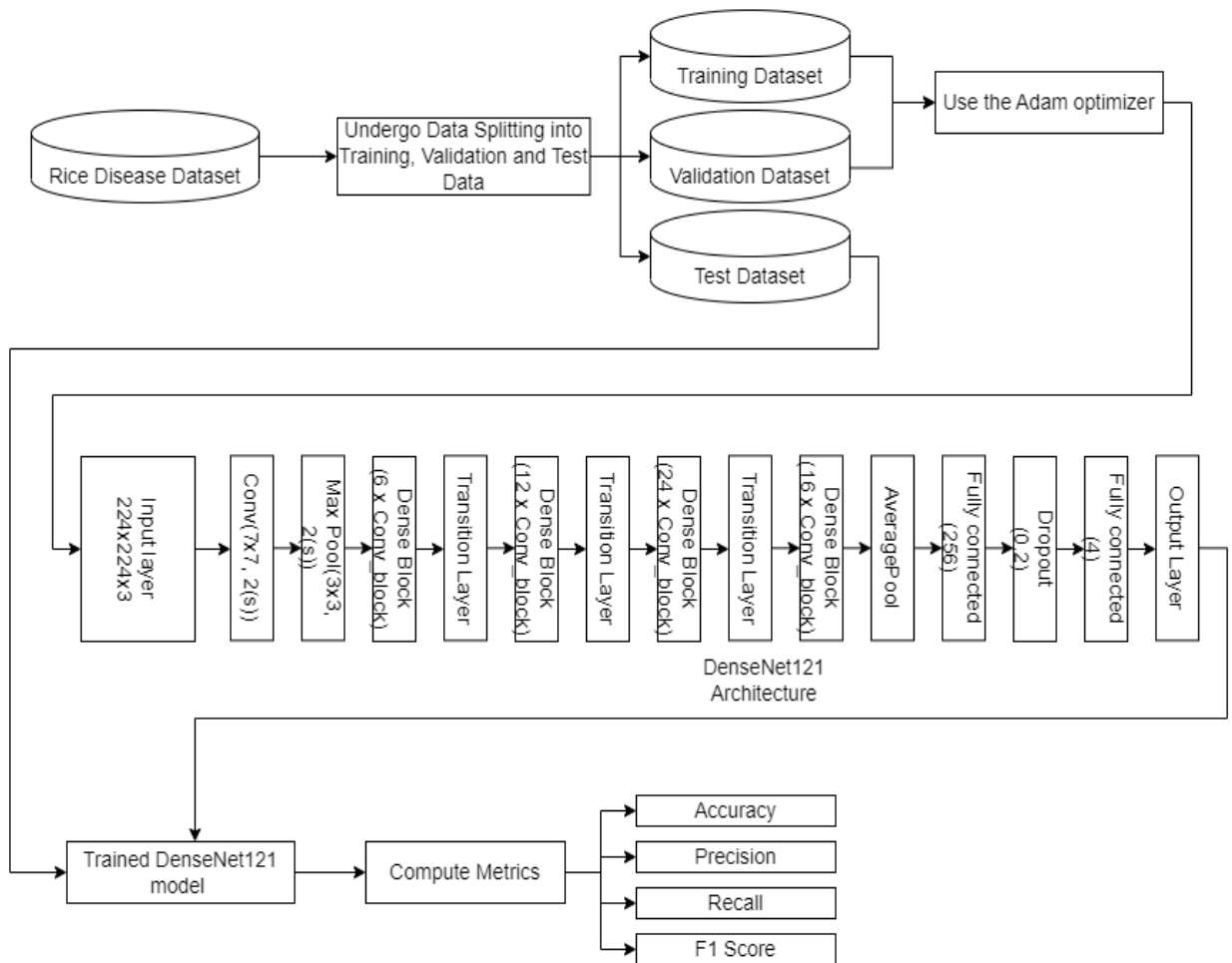
After conducting environmental scanning, the researchers have gathered necessary information to address the aims of the study. The researchers have specified all the requirements of the project to fulfill its objectives and purpose. Also, to make the project more efficiently and effectively.

System Architecture

The system has planned to develop a mobile application that requires users to have an android device in which they can interact the application. The Figure 8 illustrates the input, process, and output of the system (a) and for the model algorithm and to be computed by its performance metrics (b).



(a) Input Process Output



(b) Simulation Steps used in this Study

Figure 8. Conceptual Framework of the Study

Functional and Non-functional Requirements

The following are the functional and non-functional requirements of the system that are expected to perform and focus on users' expectation.

Functional Requirements

1. The system must be able to identify common diseases affecting crops from input images.
2. The system must be able to classify healthy and diseased rice plant evaluated by its performance metrics.
3. The system must provide information about the disease and its estimated damages to the farmers and stakeholders, and predict farmer's harvest or production output by using data mining techniques.
4. The system must be able to provide treatments prescribed by the experts and prevention strategies to farmers.

Non-functional Requirements

1. The system must be easy to use and intuitive for farmers and other stakeholders with limited technical expertise.
2. The system must be able to sync in online and offline mode.
3. Compatible with Android devices to run the application
4. The system must be reliable and maintainable, with a low risk of errors and downtime.

Use Case Diagram

A Use Case Diagram is a graphical model that emphasizes the users' interaction. The user of the proposed system has actions that are highlighted in the figure. Figure 9 highlights the details of system users' and interactions within the system.

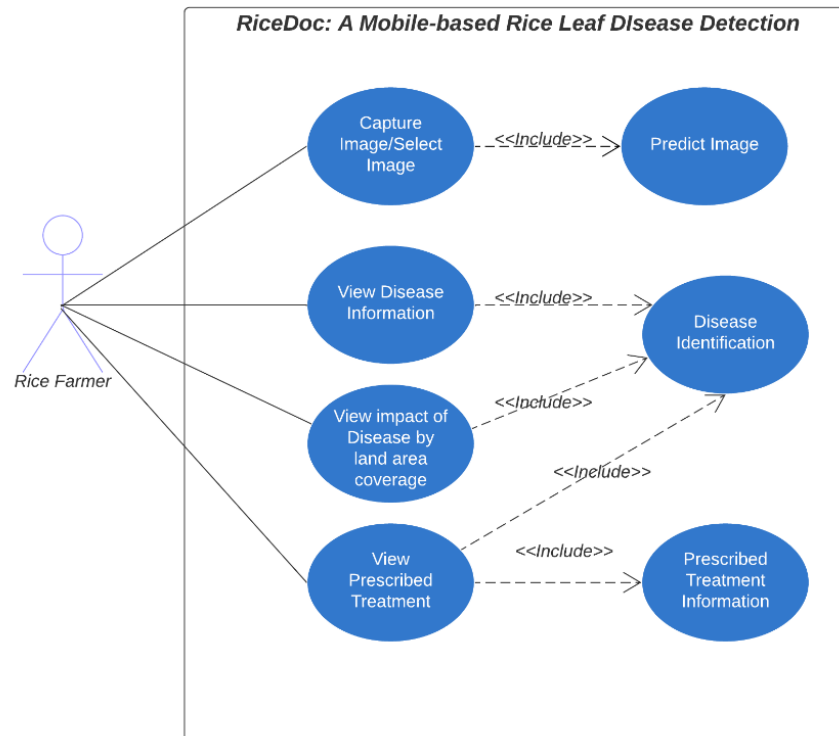


Figure 9. Use Case Diagram of the System

Context Flow Diagram

The researchers have used the Context Flow Diagram (CFD) in order to understand the flow of actions and interactions of the system. Figure 10 highlights the context flow diagram of the project that illustrates the flow of actions within the system.

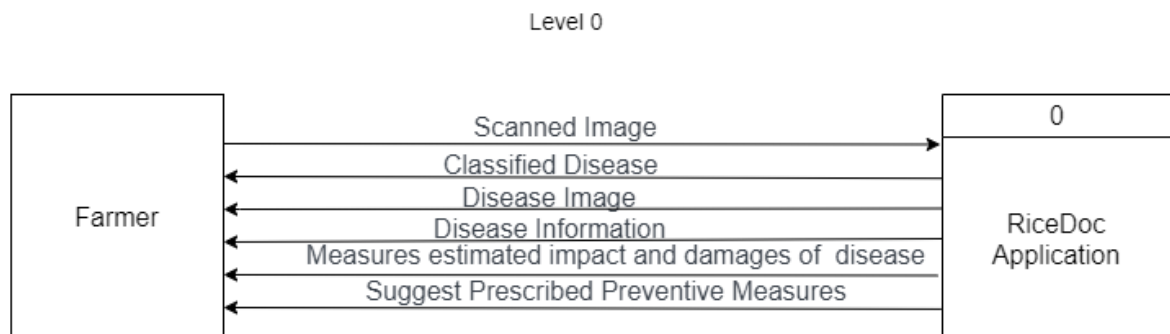


Figure 10. Context Flow Diagram of the System

Level 1 Data Flow Diagram

The researchers used the Data Flow Diagram (DFD) that is derived from Context Flow Diagram (CFD) that depicts the system's data flow in detail. In figure 11 is the data flow diagram of the system that is being broken down into sub-processes.

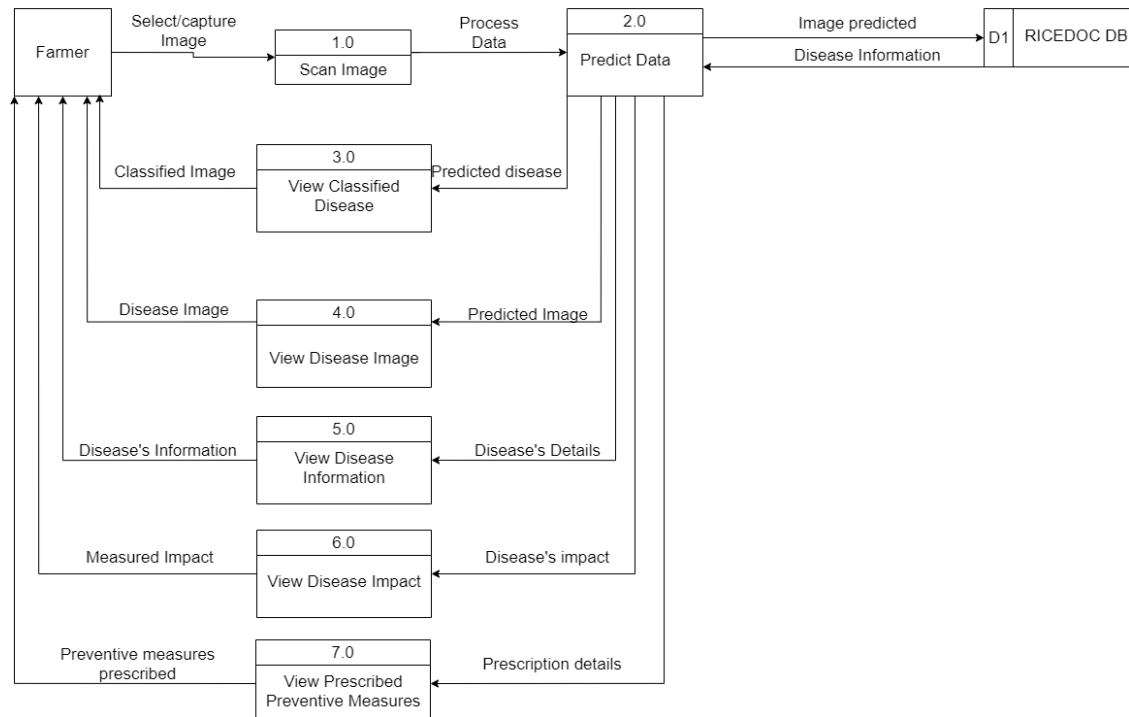


Figure 11. Data Flow Diagram of the System

System Design

This section of the document discusses the planned system designs as well as the technologies, concepts and theories to be used in this study.

Entity Relationship Diagram

The Entity Relationship Diagram illustrates the logical structures and relationship of entities within a database of the system. Figure 12 illustrates the relationship between entities of the proposed system.

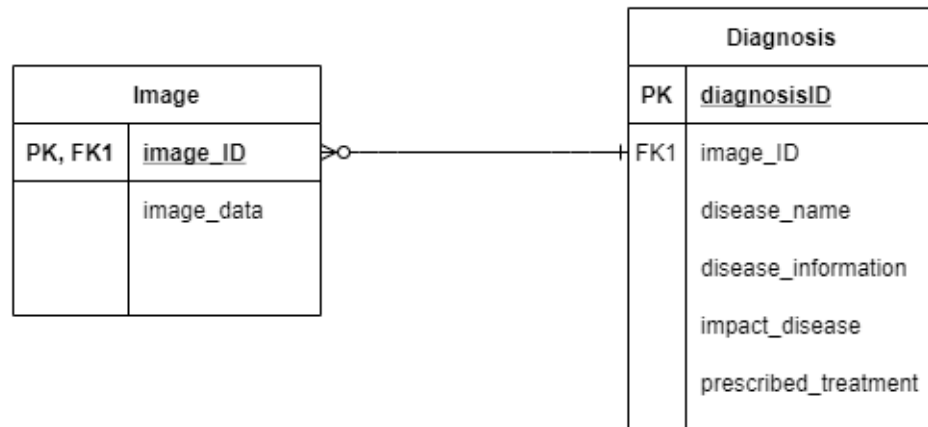


Figure 12. Entity Relationship Diagram of the System

Data Dictionary

Data dictionary provides a comprehensive and detailed overview of the data elements used in the database system, including names, data types and attributes. Table 3 and Table 4 presents the data dictionary of the database.

Table 3. Image

Attributes	Data Type	Key Type	Description
image_ID	int (10)	Primary Key	Unique identifier for an image
image_disease	BLOB		Content of the image

Table 4. Diagnosis

Attributes	Data Type	Key Type	Description
diagnosisID	int (10)	Primary Key	Unique identifier for a diagnosis
image_ID	int (10)	Foreign Key	Unique identifier for an image
diseases_name	varchar (100)		Name of the diagnosed disease
disease_information	varchar (500)		Details of the disease

prescribed_treatment	varchar (300)		Recommended treatment for the disease
impact_disease	varchar (300)		Consequence of the disease

Technologies, Concepts and Theories

The following were the technologies, concepts and theories applied by the researchers to develop the project.

Android Studio

Android Studio is the official integrated development environment (IDE) for Android application development. It is based on IntelliJ IDEA, a Java integrated development environment for software, and incorporates its code editing and developer tools. Applications built in Android Studio are then compiled into the APK format for submission to the Google Play Store. Figure 13 shows the Android Studio IDE.

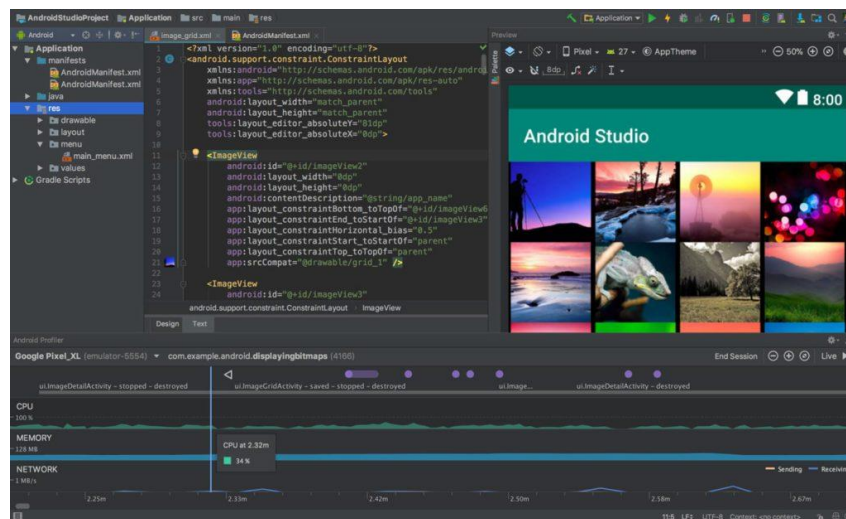


Figure 13. Android Studio IDE

DenseNet121

DenseNet121 is a convolutional neural network (CNN) architecture that belongs to the family of DenseNet models. The DenseNet121 architecture

consistently performed very well on the benchmarking image classification datasets, like ImageNet, sometimes achieving high accuracy with comparatively low parameters in comparison to different architectures. It is especially useful for tasks requiring the extraction and handling of complex features on small datasets, because it has a high level of connectivity as well as an efficient feature reuse. Figure 14 shows the DenseNet121's architecture.

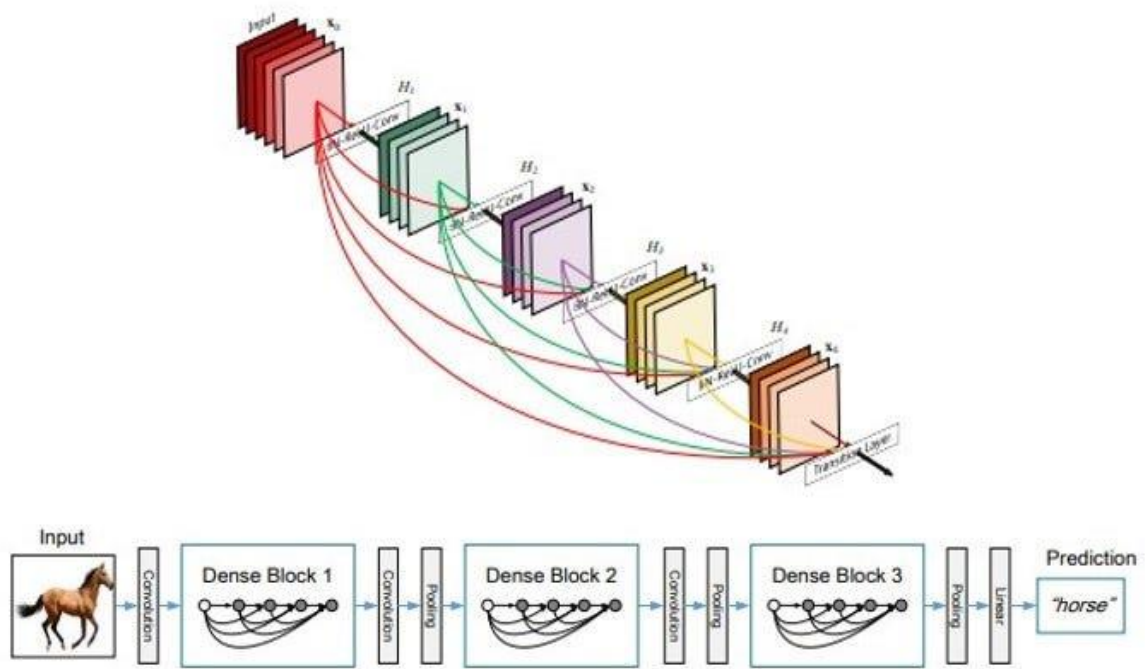


Figure 14. DenseNet121 Architecture Diagram

EfficientNetV2B1

EfficientNetV2B1 has been specifically designed with a focus on resource constrained environments, and it offers excellent tradeoffs from model size to accuracy. It is suitable for deployment on devices with limited computational resources because of its competitive performance in various computer vision tasks, such as image classification, object detection, and semantic segmentation. Figure 15 shows the architecture of EfficientNetV2B1.

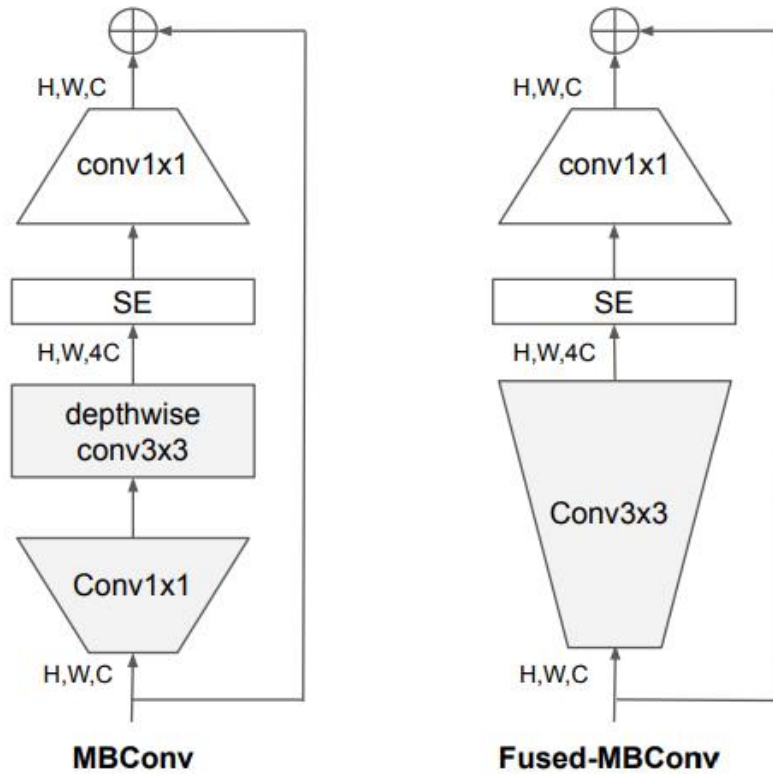


Figure 15. EfficientNetV2B1 Architecture Diagram

MobileNetV2

MobileNetV2 is a convolutional neural network (CNN) architecture specifically designed for efficient mobile vision applications. With regard to various computer vision tasks, including image classification, object detection and Semantic Clustering, MobileNETV2 has been able to achieve an extremely high level of computing efficiency as well as the best performance on that task. It is a popular choice for on device learning applications because of its low weight, in view of the fact that they are not capable of using sufficient compute resources. Figure 16 is the MobileNetV2's architecture in cnn.

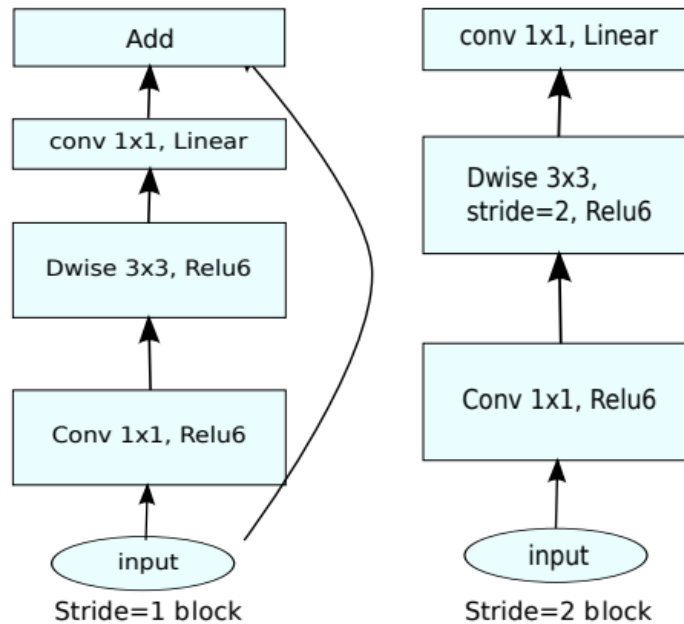


Figure 16. MobileNetV2 Architecture Diagram

Jupyter Notebook

The Jupyter Notebook is the original web application for creating and sharing computational documents. It offers a simple, streamlined, document-centric experience. Jupyter Notebook is an interactive computing environment that allows you to create and share documents containing live code, equations, visualizations, and explanatory text. Figure 17 shows the Jupyter Notebook environment.

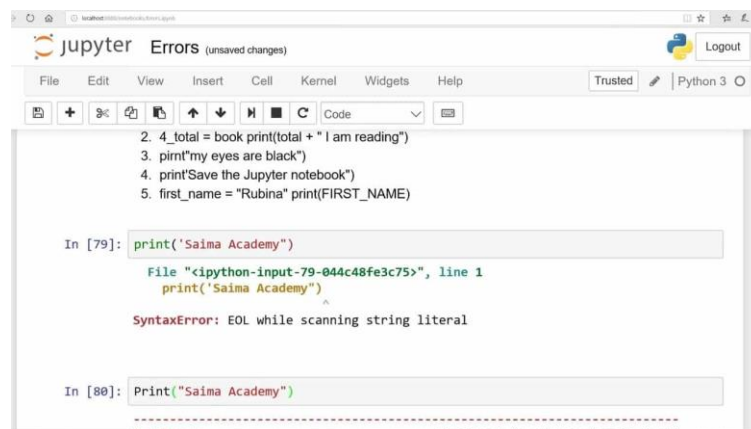


Figure 17. Jupyter Notebook IDE

Keras

Keras is an open-source software library that provides a Python interface for artificial neural networks. Keras acts as an interface for the TensorFlow library. Keras is also a high-level, deep learning API developed by Google for implementing neural networks which has the basic building blocks of neural networks. Figure 18 illustrates the keras library.

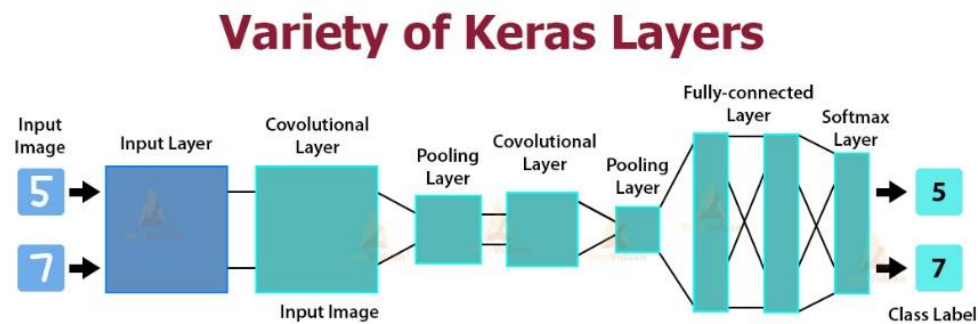


Figure 18. Keras Layers illustration

Tensorflow Library

TensorFlow is a free and open-source software library for machine learning and artificial intelligence. It can be used across a range of tasks but has a particular focus on training and inference of deep neural networks. TensorFlow is a Python library for fast numerical computing created and released by Google. Figure 13 shows the Tensorflow being instantiated.

```
File Edit Selection View Go Debug Terminal Help
testipynb x
[1] ▶ MI
import tensorflow as tf
print('tensorflow version', tf.__version__)
tensorflow version 2.0.0
[2] ▶ MI
x = [[3.]]
y = [[4.]]
print('Result: {}'.format(tf.matmul(x, y)))
Result: [[12.]]
TechBrij.com
```

Figure 19. Tensorflow Library code initiation

System Testing and Implementation

This section of the document is presenting the phase of system testing and implementation of the image classification architectures. The researchers have prospected and tested three architectures for image classification model and generate performance metrics before implementing the cnn architecture of the model. The best cnn architecture that researchers have used is DenseNet121

Table 4. MobileNetV2 Results using Adam Optimizer

	Epochs	Accuracy	Recall	Precision	F1-Score
MobileNetV2	20	95.56%	95.56	95.55	95.53
	40	95.56%	95.56	95.55	95.53
	60	96.11%	96.11	96.13	96.08
	100	95.56%	95.56	95.62	95.53
Average:		95.70%	95.70	95.71	95.67
Total Average:				95.70%	

Table 5. MobileNetV2 Results using RMSProp Optimizer

	Epochs	Accuracy	Recall	Precision	F1-Score
MobileNetV2	20	95.00%	95.00	95.22	95.01
	40	95.00%	95.00	94.99	94.98
	60	96.11%	96.11	96.13	96.08
	100	94.44%	94.44	94.52	94.41
Average:		95.14%	95.14	95.20	95.12
Total Average:				95.15%	

Table 6. EfficientNetV2B1 Results using Adam Optimizer

	Epochs	Accuracy	Recall	Precision	F1-Score
EfficientNetV2B1	20	95.56%	95.56	95.94	95.44
	40	95.56%	95.56	95.92	95.43
	60	96.67%	96.67	96.88	96.60
	100	97.22%	97.22	97.37	97.18
Average:		96.25%	95.25%	96.52%	96.16%
Total Average:				96.05%	

Table 7. EfficientNetV2B1 Results using RMSProp Optimizer

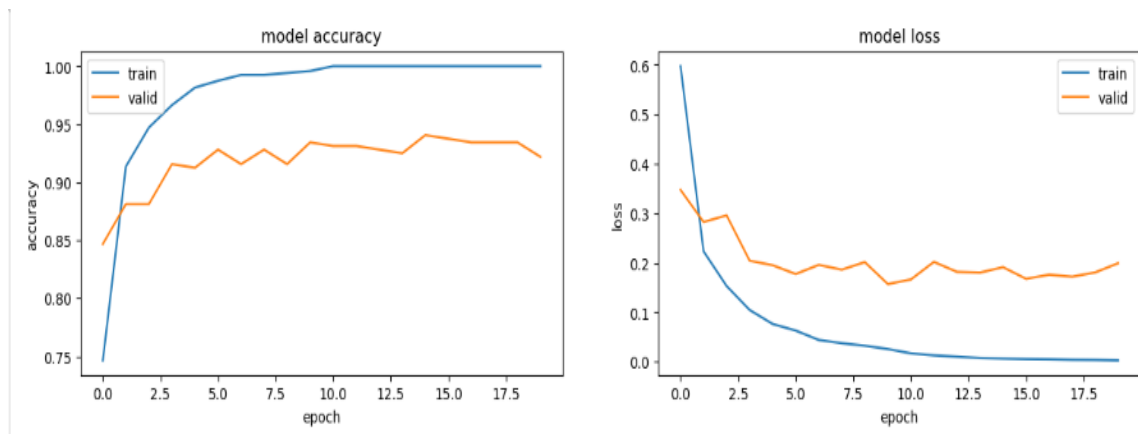
	Epochs	Accuracy	Recall	Precision	F1-Score
EfficientNetV2B1	20	95.00%	95.00	95.46	94.84
	40	96.67%	96.67	96.88	96.60
	60	97.22%	97.22	97.37	97.18
	100	97.22%	97.22	97.37	97.18
	Average:	96.53%	96.53%	96.77%	96.45%
Total Average:				96.57%	

Table 8. DenseNet121 Results using Adam Optimizer

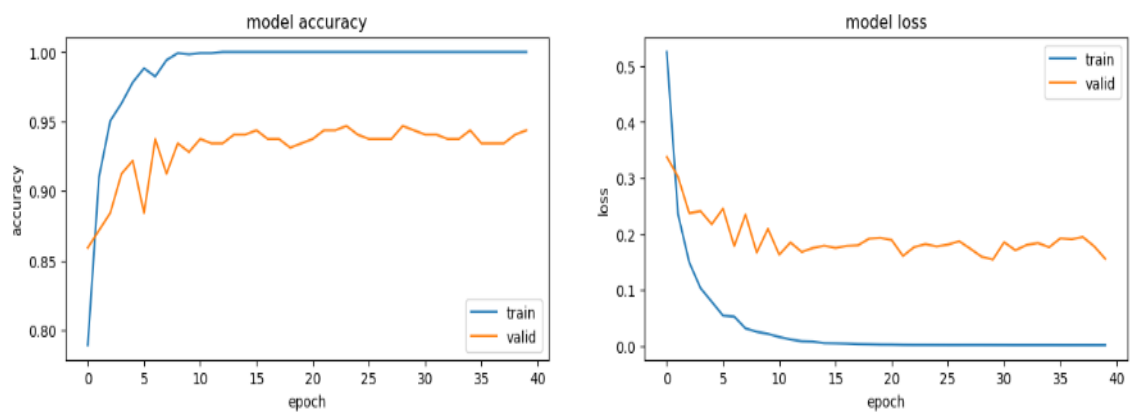
	Epochs	Accuracy	Recall	Precision	F1-Score
DenseNet121	20	96.67%	95.67	96.69	96.65
	40	98.33%	98.33	98.36	98.84
	60	97.22%	97.22	97.25	97.23
	100	97.22%	97.22	97.35	97.25
	Average:	97.36%	97.36%	97.41%	97.25%
Total Average:				97.38%	

Table 9. DenseNet121 Results using RMSProp Optimizer

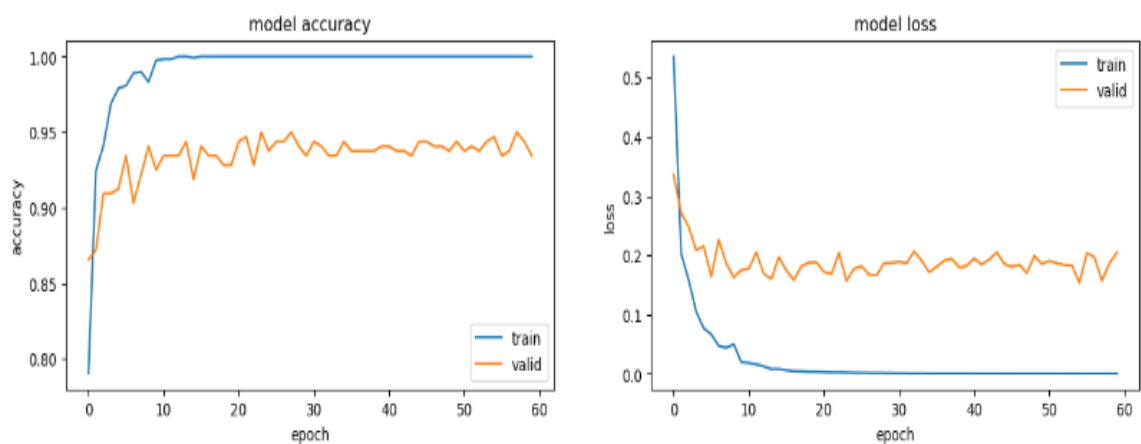
	Epochs	Accuracy	Recall	Precision	F1-Score
DenseNet121	20	97.22%	97.22	97.25	97.23
	40	97.22%	97.22	97.35	97.25
	60	97.78%	97.78	97.84	97.79
	100	97.78%	97.78	97.84	97.79
	Average:	97.5%	97.5%	97.84%	97.79%
Total Average:				97.27%	



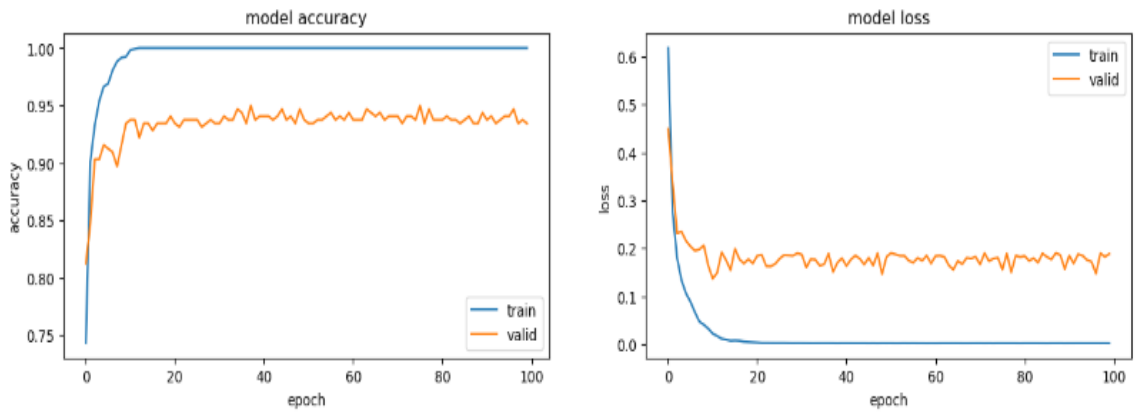
(a) MobileNetV2 Adam Optimizer 20 epochs



(b) MobileNetV2 Adam Optimizer 40 epochs



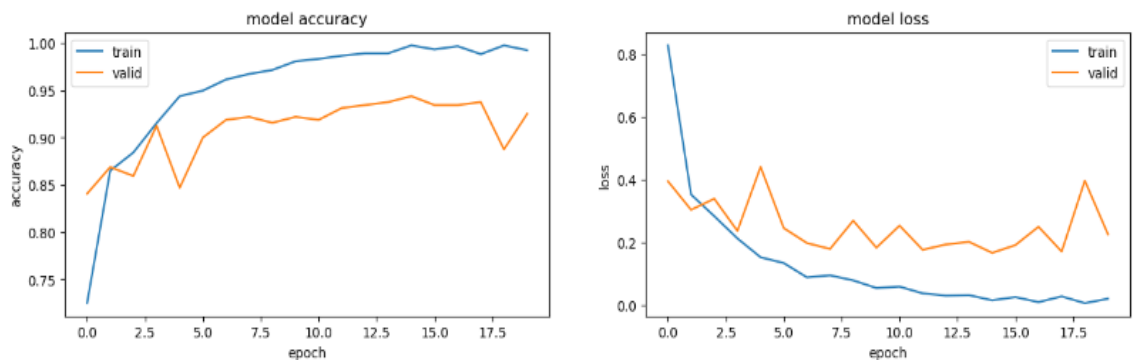
(c) MobileNetV2 Adam Optimizer 60 epochs



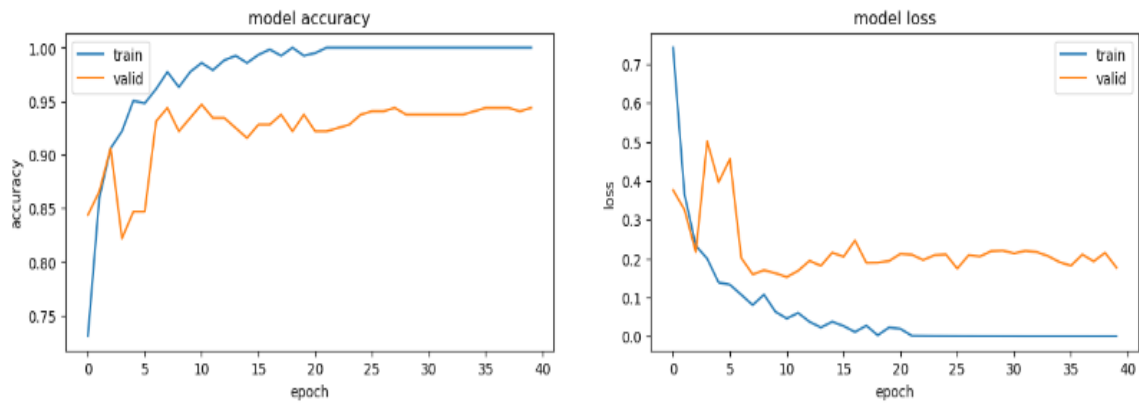
(d) MobileNetV2 Adam Optimizer 100 epochs

Figure 20. MobileNetV2 using Adam Optimizer Model Accuracy and Loss

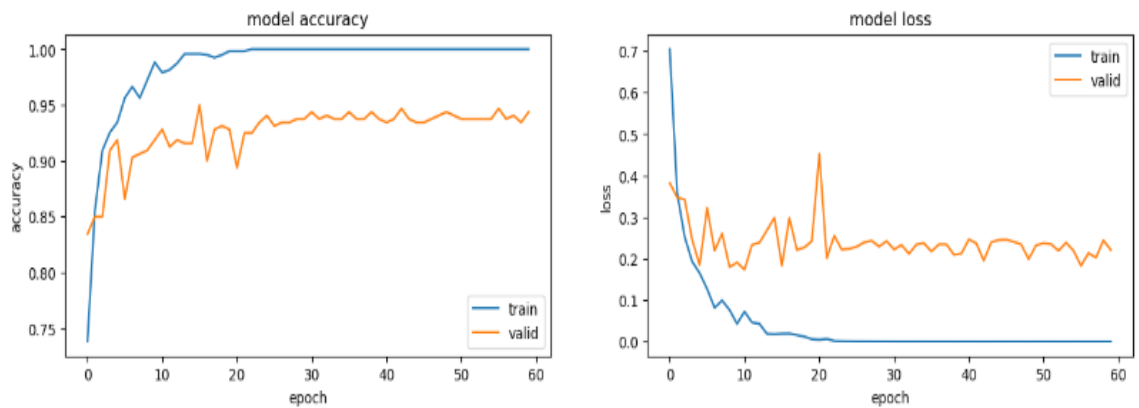
Figure 20 (a) illustrates the model accuracy and loss of the MobileNetV2 architecture generated during the training of the architecture. The architecture was trained under 20 epochs using Adam optimizer as one of the hyper parameters used during training. Figure 20 (b) illustrates the model accuracy and loss generated during the training of the architecture it was trained on 40 epochs and have the same optimizer used previously. Figure 20 (c) also a model accuracy and loss of the trained architecture, it was trained on 60 epochs. Lastly, figure 20 (d) is trained under 100 epochs using the same optimizer. The performance of this architecture with its optimizer used has been tabulated in Table 4.



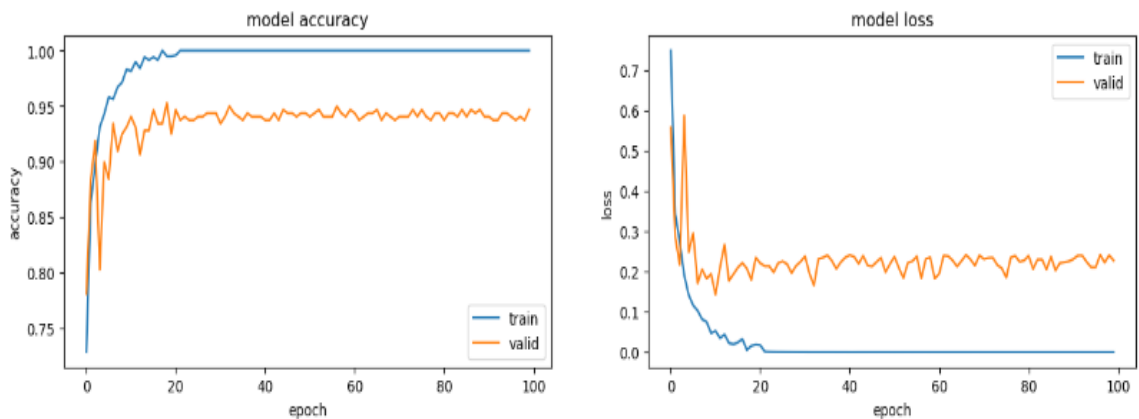
(a) MobileNetV2 RMSProp Optimizer 20 epochs



(b) MobileNetV2 RMSProp Optimizer 40 epochs



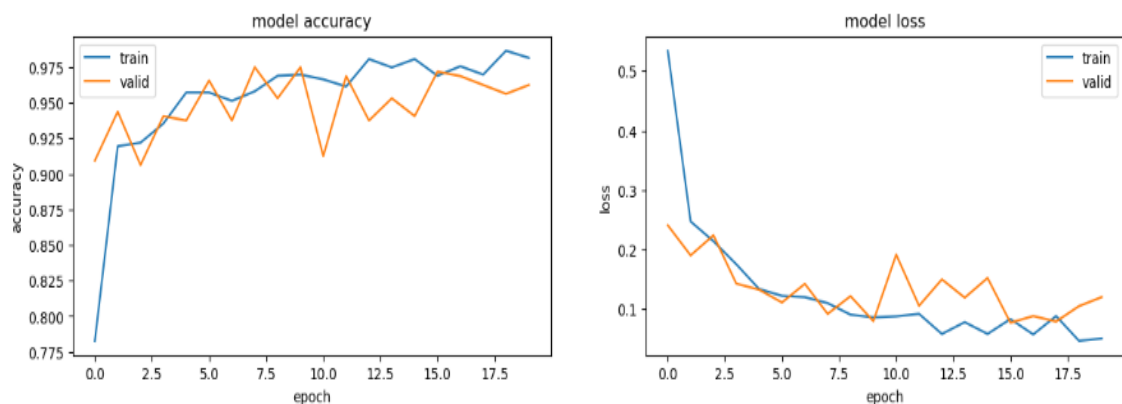
(c) MobileNetV2 RMSProp Optimizer 60 epochs



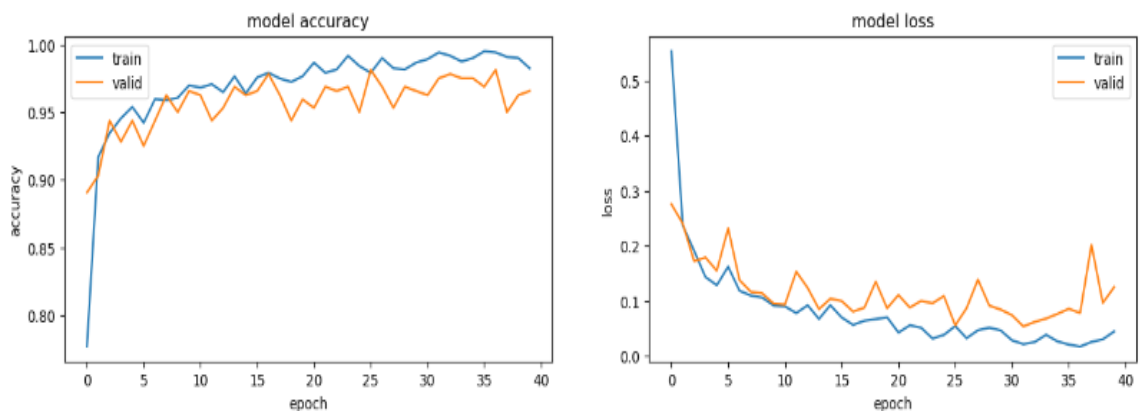
(d) MobileNetV2 RMSProp Optimizer 100 epochs

Figure 21. MobileNetV2 using RMSProp Optimizer Model Accuracy and Loss

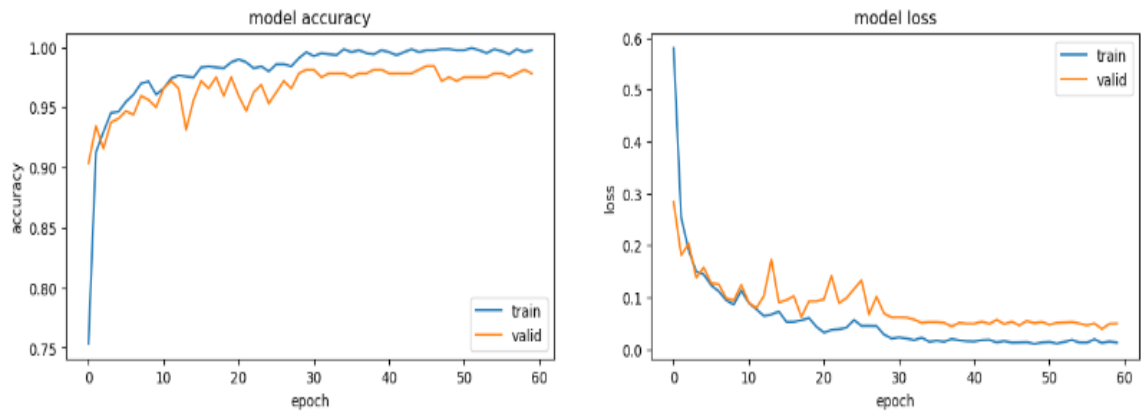
Figure 21 (a) illustrates the model accuracy and loss of the MobileNetV2 architecture generated during the training of the architecture. The architecture was trained under 20 epochs using RMSProp optimizer as one of the hyper parameters used during training. Figure 21 (b) illustrates the model accuracy and loss generated during the training of the architecture it was trained on 40 epochs and have the same optimizer used previously. Figure 21 (c) also a model accuracy and loss of the trained architecture, it was trained on 60 epochs. Lastly, figure 21 (d) is trained under 100 epochs using the same optimizer. The performance of this architecture with its optimizer used has been tabulated in Table 5.



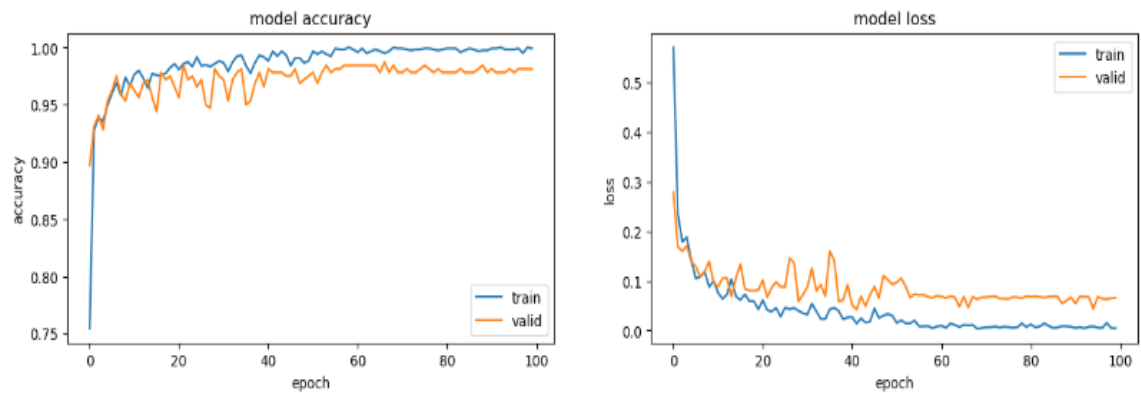
(a) EfficientNetV2B1 Adam Optimizer 20 epochs



(b) EfficientNetV2B1 Adam Optimizer 40 epochs



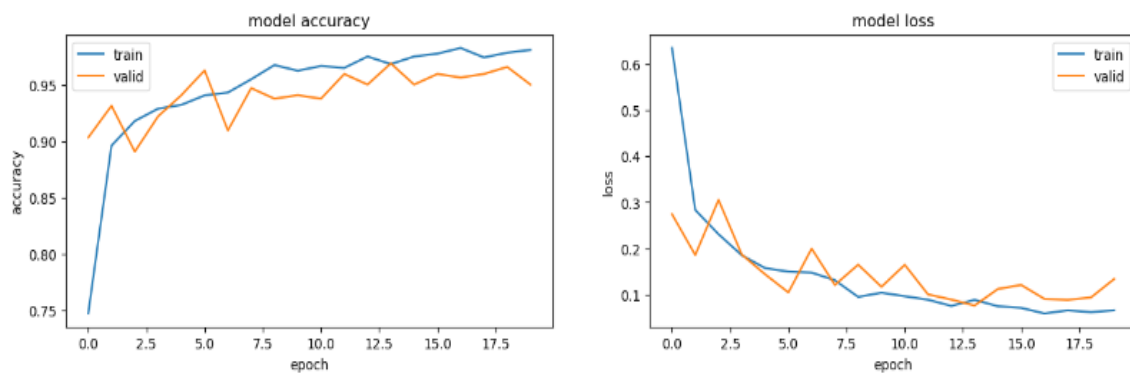
(c) EfficientNetV2B1 Adam Optimizer 60 epochs



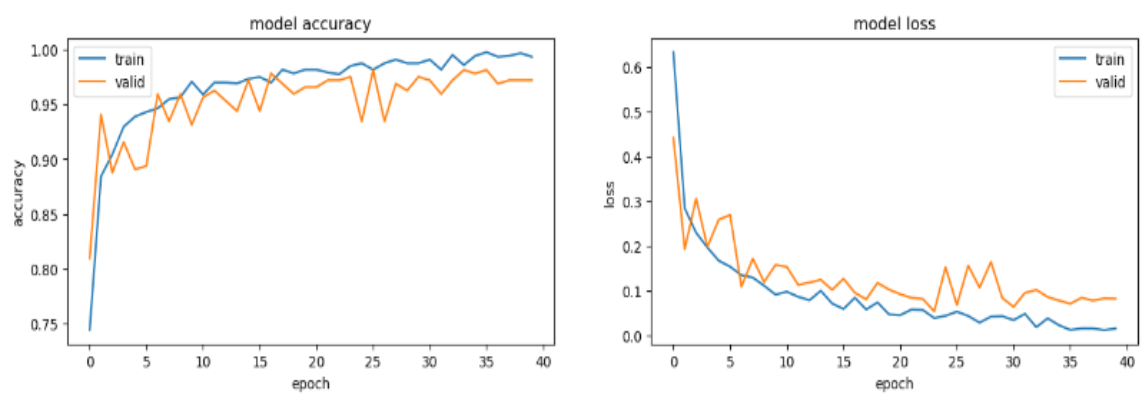
(d) EfficientNetV2B1 Adam Optimizer 100 epochs

Figure 22. EfficientNetV2B1 using Adam Optimizer Model Accuracy and Loss

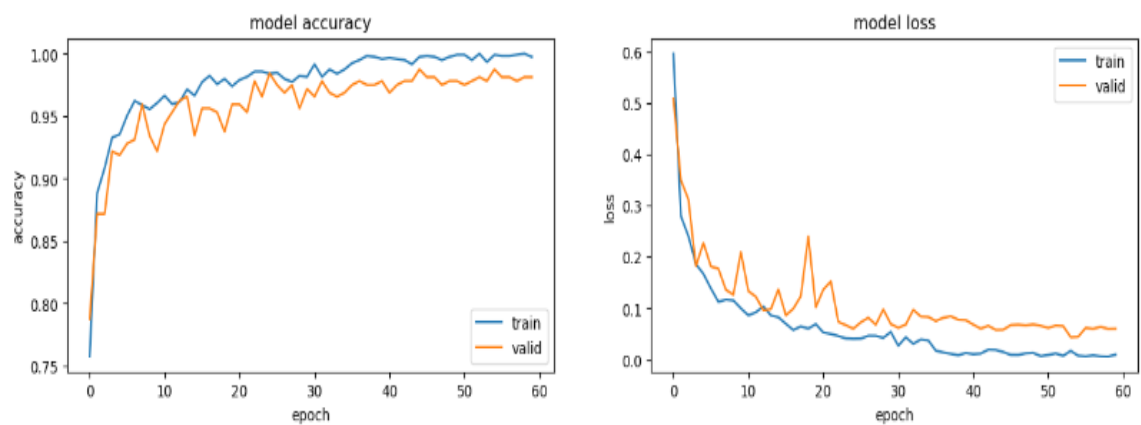
Figure 22 (a) illustrates the model accuracy and loss of the EfficientNetV2B1 architecture generated during the training of the architecture. The architecture was trained under 20 epochs using Adam optimizer as one of the hyper parameters used during training. Figure 22 (b) illustrates the model accuracy and loss generated during the training of the architecture it was trained on 40 epochs and have the same optimizer used previously. Figure 22 (c) also a model accuracy and loss of the trained architecture, it was trained on 60 epochs. Lastly, figure 22 (d) is trained under 100 epochs using the same optimizer. The performance of this architecture with its optimizer used has been tabulated in Table 6.



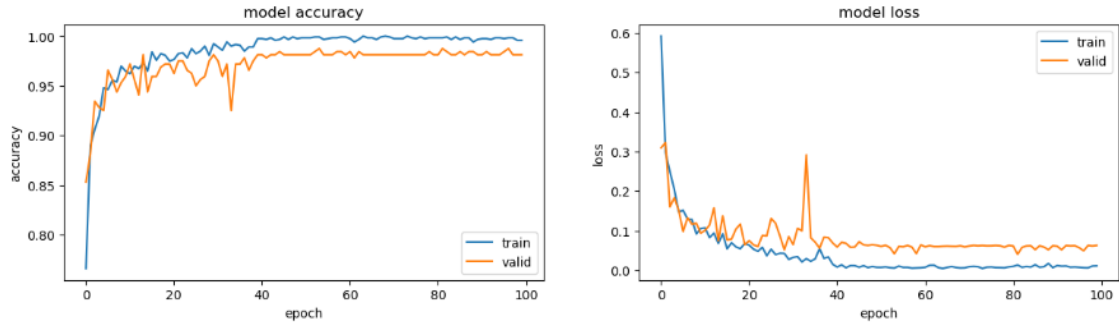
(a) EfficientNetV2B1 RMSProp Optimizer 20 epochs



(b) EfficientNetV2B1 RMSProp Optimizer 40 epochs



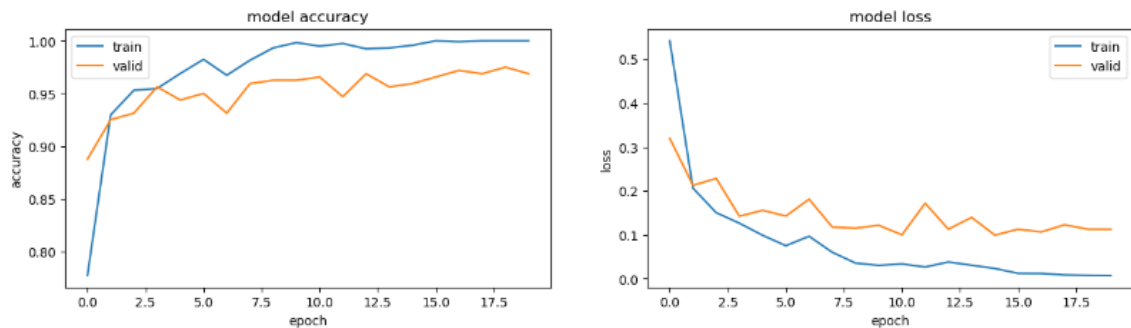
(c) EfficientNetV2B1 RMSProp Optimizer 60 epochs



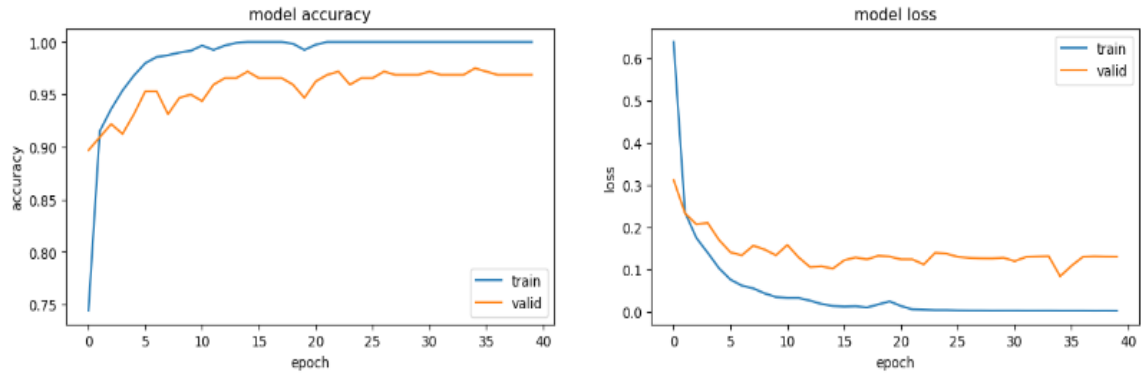
(d) EfficientNetV2B1 RMSProp Optimizer 100 epochs

Figure 23. *EfficientNetV2B1 using RMSProp Optimizer Model Accuracy and Loss*

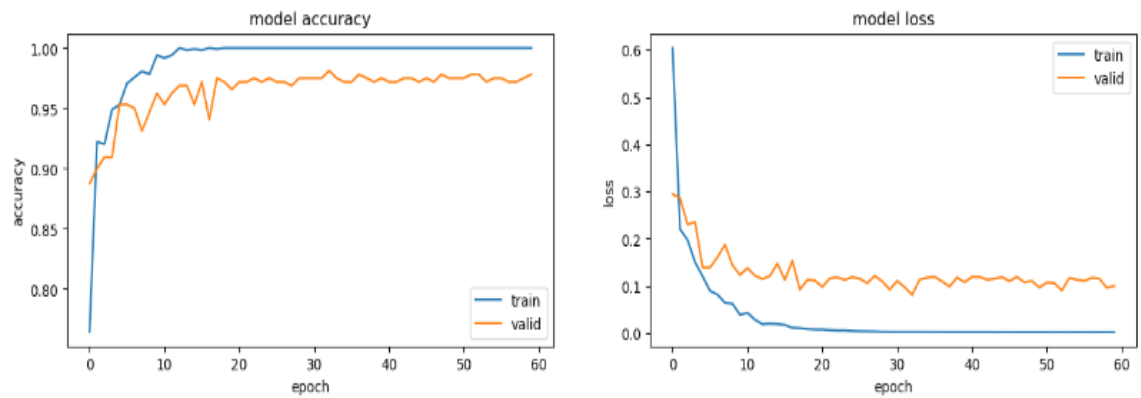
Figure 23 (a) illustrates the model accuracy and loss of the EfficientNetV2B1 architecture generated during the training of the architecture. The architecture was trained under 20 epochs using RMSProp optimizer as one of the hyper parameters used during training. Figure 23 (b) illustrates the model accuracy and loss generated during the training of the architecture it was trained on 40 epochs and have the same optimizer used previously. Figure 23 (c) also a model accuracy and loss of the trained architecture, it was trained on 60 epochs. Lastly, figure 23 (d) is trained under 100 epochs using the same optimizer. The performance of this architecture with its optimizer used has been tabulated in Table 7.



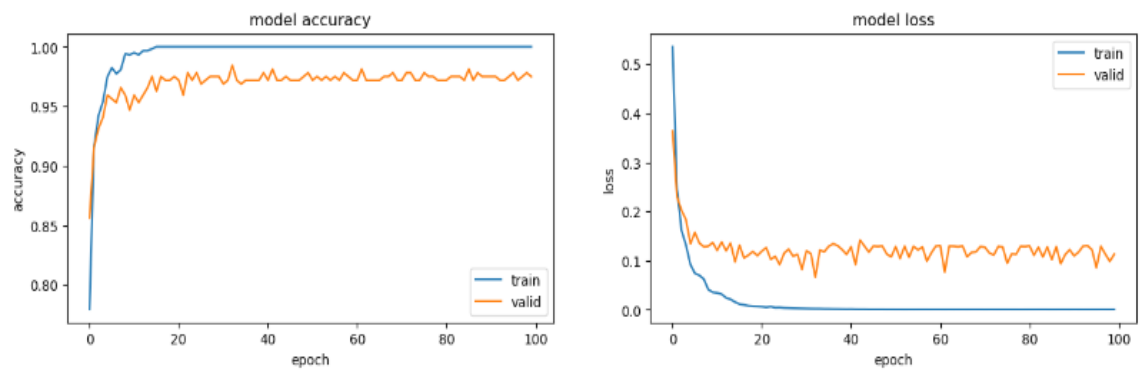
(a) DenseNet121 Adam Optimizer 20 epochs



(b) DenseNet121 Adam Optimizer 40 epochs



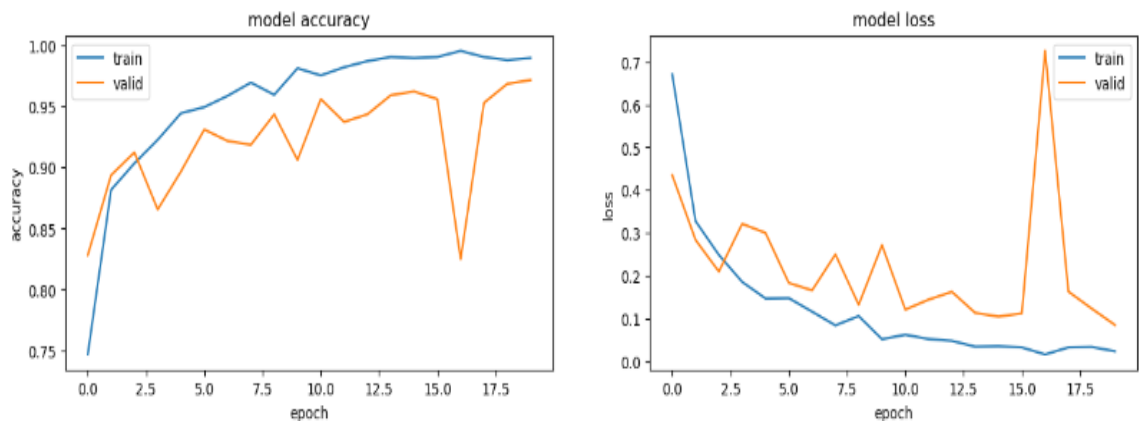
(c) DenseNet121 Adam Optimizer 60 epochs



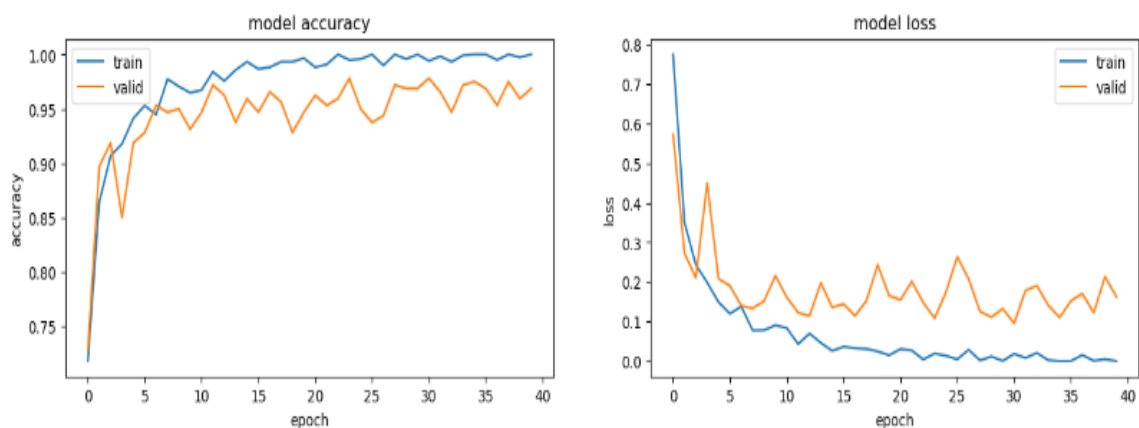
(d) DenseNet121 Adam Optimizer 100 epochs

Figure 24. DenseNet121 using Adam Optimizer Model Accuracy and Loss

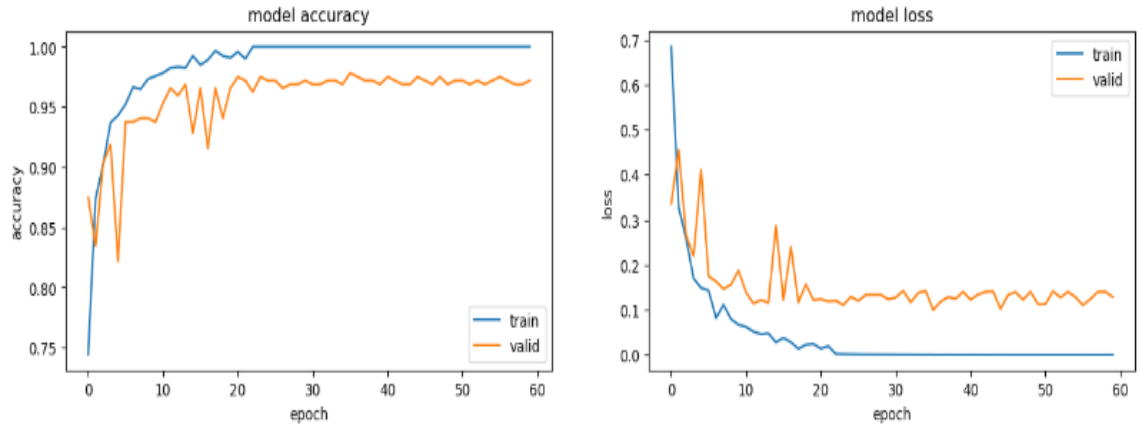
Figure 24 (a) illustrates the model accuracy and loss of the DenseNet121 architecture generated during the training of the architecture. The architecture was trained under 20 epochs using Adam optimizer as one of the hyper parameters used during training. Figure 24 (b) illustrates the model accuracy and loss generated during the training of the architecture it was trained on 40 epochs and have the same optimizer used previously. Figure 24 (c) also a model accuracy and loss of the trained architecture, it was trained on 60 epochs. Lastly, figure 24 (d) is trained under 100 epochs using the same optimizer. The performance of this architecture with its optimizer used has been tabulated in Table 8.



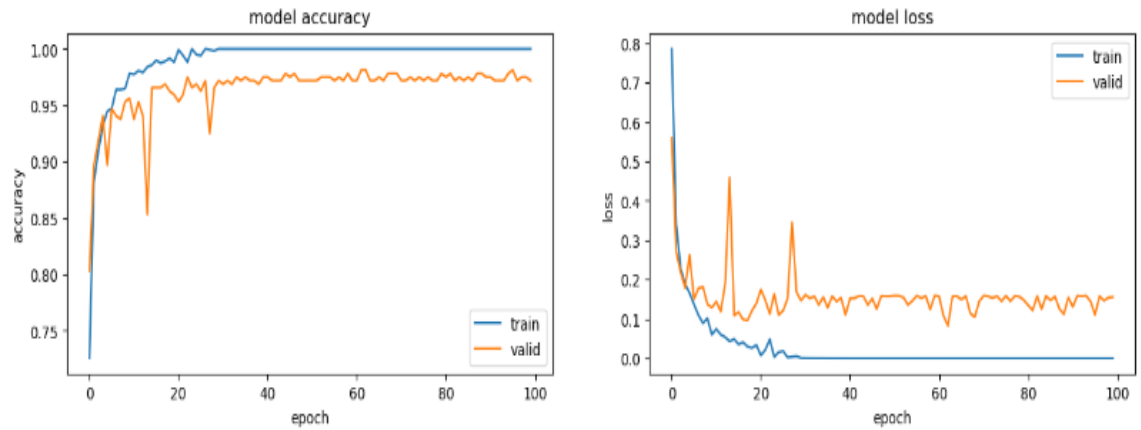
(a) DenseNet121 RMSProp Optimizer 20 epochs



(b) DenseNet121 RMSProp Optimizer 40 epochs



(c) DenseNet121 RMSProp 60 epochs



(d) DenseNet121 RMSProp 100 epochs

Figure 25. DenseNet121 using RMSProp Optimizer Model Accuracy and Loss

Figure 25 (a) illustrates the model accuracy and loss of the DenseNet121 architecture generated during the training of the architecture. The architecture was trained under 20 epochs using RMSProp optimizer as one of the hyper parameters used during training. Figure 25 (b) illustrates the model accuracy and loss generated during the training of the architecture it was trained on 40 epochs and have the same optimizer used previously. Figure 25 (c) also a model accuracy and loss of the trained architecture, it was trained on 60 epochs. Lastly, figure 25 (d) is

trained under 100 epochs using the same optimizer. The performance of this architecture with its optimizer used has been tabulated in Table 9.

Table 10. Summary of Dataset

Image Partition: 70-20-10				
Image Label	No. Images	Train	Validation	Test
Brown Spot	438	306	87	45
Leaf Blast	438	306	87	45
Healthy	438	306	87	45
Sheath Blight	438	306	87	45
Total:	1752	1224	348	180

Table 11. Summary of Hyper parameters used

Parameters	Value
Optimizer	Adam and RMSProp
Learning Rate	0.001
Learning Rate Scheduler	ReduceOnPlateau
Patience	5
Factors	0.2
min_lr	1.000E-07

Overall, the three cnn architecture that has been used to evaluate that to be implemented in the model. The researchers have decided to implement the DenseNet121 cnn architecture with Adam optimizer as one of the hyper parameters used, to become the classification model of the system.

System Maintenance

This section of the document discusses the maintenance of the system. Maintaining a mobile classification application and its model architecture requires an extensive effort including the developers and end-users. The implementation of

the cnn architecture particularly the DenseNet121 has been approved by the panel of examiners.

System Maintenance Plan

The researchers will set a series of maintenance to examine if there's any bugs encountered during the testing phase of the system and to ensure the user's experience while using the mobile app. The proposed mobile application of the researchers is going to be launched as an offline application, researchers cannot guarantee if there's an internet connection base on user's location.

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